

8

St George Girls' High School

Trial Higher School Certificate Examination

17 minutes
a question.

2003



Mathematics

Extension 1

Total Marks – 84

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Attempt ALL questions.
- Begin each question on a new page
- Write your student number on each page
- All necessary working must be shown.
- Diagrams are not to scale.
- Board-approved calculators may be used.
- The mark allocated for each question is listed at the side of the question.

Question	Mark
Q1	/12
Q2	/12
Q3	/12
Q4	/12
Q5	/12
Q6	/12
Q7	/12
Total	/84

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Question 1 – (12 marks) – Start a new page

Marks

a) Find the exact value of $\int_2^3 \frac{x^2}{x^3 - 7} dx$ 2

b) Solve for x : $\frac{2}{x-1} \leq 1$ 3

c) $P(19, -15)$ is the point which divides the line interval 'AB' externally in the ratio 3:2.
Find the coordinates of $B(x, y)$ given $A(-2, 3)$. 3

d) (i) Find $\frac{d}{dx}(\tan^{-1}x + x)$

(ii) Hence, evaluate $\int_0^1 \frac{x^2 + 2}{x^2 + 1} dx$ 4

(leave in exact form).

Question 2 – (12 marks) – Start a new page

Marks

- a) The equation $x^3 - mx + 2 = 0$ has two of its roots equal. 4
- (i) Write down expressions for the sum of the roots and for the product of the roots.
- (ii) Hence, find the value of m .
- b) The polynomial equation $8x^3 - 36x^2 + 22x + 21 = 0$ has roots which form an arithmetic progression. 3
- Find the roots of the polynomial.
- c) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola with the equation $x^2 = 4ay$.
- It is given that the chord PQ has equation $y = \left(\frac{p+q}{2}\right)x - apq$ 3
- (i) Show that the gradient of the tangent at P is p .
- (ii) Prove that if PQ passes through the focus, then the tangent at P is parallel to the normal at Q .
- d) (i) Write down the equation for the inverse function of $y = 2^x$, write your response with y as the subject. 2
- (ii) Write down the domain of the inverse function from part (i).

Question 3 – (12 marks) – Start a new page

Marks

a) Find the term independent of x in the expansion of $\left(x - \frac{2}{x^3}\right)^{12}$ 3

b) Find the greatest coefficient in the expansion of $(2 + 3x)^{14}$ 4

c) (i) Show that $\sqrt{12} \sin x + 2 \cos x \equiv 4 \cos\left(x - \frac{\pi}{3}\right)$ 5

(ii) Hence, solve the equation $\sqrt{12} \sin x + 2 \cos x = -2\sqrt{2}$ for $0 \leq x \leq 2\pi$

[Give all answers correct to two decimal places]

Question 4 – (12 marks) – Start a new page

Marks

- a) The region bounded by the curve $y = \sin x$, the x -axis and the lines $x = \frac{\pi}{12}$ and $x = \frac{\pi}{4}$ is rotated through one complete revolution about the x -axis. Find the volume of the solid so formed.

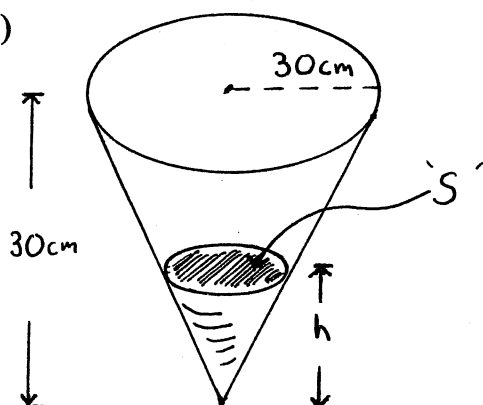
4

[Give your answer in terms of π].

- b) Use Mathematical induction to show that the expression $7^n + 5$ is divisible by 6 for all positive integers n .

4

c)



Water is poured into a conical vessel at a constant rate of 24cm^3 per second. The depth of water is h cm at any time t seconds.

4

What is the rate of increase of the area of the surface 'S' of the water when the depth is 16 cm?

[NOT TO SCALE]

Question 5 – (12 marks) – Start a new page

Marks

- a) Newton's Law of Cooling states that when an object at temperature T° is placed in an environment at a temperature of R° , then the rate of temperature loss is given by the equation

$$\frac{dT}{dt} = k(T - R)$$

where t is the time in seconds and k is a constant.

- (i) Show that $T = R + Ae^{kt}$ is a solution to the equation. 1

- (ii) A packet of peas, initially at 24°C is placed in a snap-freeze refrigerator in which the internal temperature is maintained at -40°C .

After 5 seconds the temperature of the packet is 19°C . How long will it take for the packet's temperature to reduce to 0°C ? 3

- b) Consider the function $y = \log_e \left(\frac{2x}{2+x} \right)$

- (i) Show that the domain of the function is: $x < -2, x > 0$ 2

- (ii) Find the value of x for which $y = 0$ 1

- (iii) Show that $\frac{dy}{dx} = \frac{2}{x(2+x)}$ and hence show that the function is increasing for all x in the domain. 2

- (iv) Find any possible points of inflexion. 1

- (v) Find $\lim_{x \rightarrow \infty} \left[\log_e \left(\frac{2x}{2+x} \right) \right]$ 1

- (vi) Sketch the graph of the function. 1

Question 6 – (12 marks) – Start a new page

Marks

a) By noting that $(1+x)^n = \sum_{r=0}^n {}^nC_r x^r$,

prove that

(i) $\sum_{r=0}^n {}^nC_r = 2^n$

1

(ii) $\sum_{r=1}^n r \cdot {}^nC_r = n \cdot 2^{n-1}$

2

b) Evaluate $\int_1^3 \frac{dx}{(1+x)\sqrt{x}}$ using the substitution $u = \sqrt{x}$, give the EXACT value.

4

c) A particle moving in Simple Harmonic Motion starts from rest at a distance 10 metres to the right of its centre of oscillation O . The period of the motion is 2 seconds.

(i) Find the speed of the particle when it is 4 metres from its starting point.

5

(ii) Find the time taken by the particle to first reach the point 4 metres from its starting point, in seconds correct to two decimal points.

$$x = -10 \cos(\pi t + \alpha)$$

$$x = -10 \cos(\pi t + \alpha)$$

$$\dot{x} = 10\pi \sin \pi t$$

$$\frac{2\pi}{n} = 2$$

$$2\pi = 2n$$

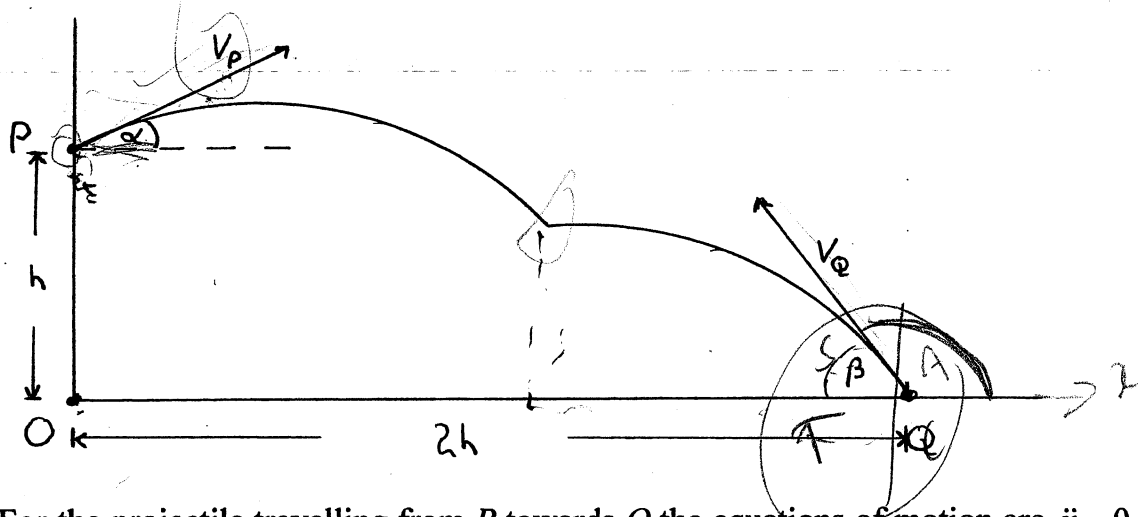
Question 7 – (12 marks) – Start a new page

Marks

- a) O and Q are two points $2h$ metres apart on horizontal ground. P is a point h metres directly above O .

6

A particle is projected from P towards Q with speed $V_P \text{ ms}^{-1}$ at an angle ' α ' above the horizontal. At the same time another particle is projected from Q towards P with speed $V_Q \text{ ms}^{-1}$ at an angle of ' β ' above the horizontal. The two particles collide ' T ' seconds after projection.



- (i) For the projectile travelling from P towards Q the equations of motion are $\ddot{x} = 0$ and $\ddot{y} = -g$.

Use calculus to show that at time t seconds, its horizontal distance x_P from O and its vertical height y_P from O are given by $x_P = (V_P \cos \alpha)t$ and

$$y_P = (V_P \sin \alpha)t - \frac{1}{2}gt^2 + h$$

- (ii) For the particle going from Q towards P , write down expressions for the horizontal distance x_Q from Q and its vertical height y_Q from Q at time t seconds.

- (iii) Hence, show that $\frac{V_P}{V_Q} = \frac{2 \sin \beta - \cos \beta}{2 \sin \alpha + \cos \alpha}$

Question 7 (cont'd)

Marks

- b) (i) Write down an expression for $\sin(x - y)$

6

- (ii) If $\sin \alpha = c$ and $\sin(60^\circ - \alpha) = d$, prove that $c^2 + cd + d^2 = \frac{3}{4}$

- (iii) If $\triangle ABC$ is equilateral and D is any point on the side BC and if a and b are the lengths of the perpendiculars from D to AB and AC respectively, prove

$$AD = \frac{2}{\sqrt{3}} \sqrt{a^2 + ab + b^2}$$

EXTENSION 1 - SOLUTIONS.

QUESTION 1: (12 MARKS)

$$\begin{aligned} \text{(a)} \quad \int_2^3 \frac{x^2}{x^3-7} dx &= \frac{1}{3} \left[\ln(x^3-7) \right]_2^3 \\ &= \frac{1}{3} (\ln 20 - \ln 1) \\ &= \frac{1}{3} \ln 20 \end{aligned}$$

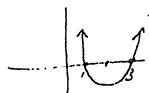
$$\text{(b)} \quad \frac{2}{x-1} \leq 1 \quad x \neq 1$$

$$\begin{aligned} 2(x-1) &\leq (x-1)^2 \\ 0 &\leq (x-1)^2 - 2(x-1) \\ 0 &\leq (x-1)(x-3) \end{aligned}$$

$$\Rightarrow x \leq 1 \text{ OR } x \geq 3$$

but $x \neq 1$

$$\therefore x < 1 \text{ OR } x \geq 3$$



$$\text{(c)} \quad \frac{A(-2, 3)}{3: -2} \quad \frac{B(x, y)}{3: -2}$$

$$\text{Hence } (19, -15) = \left(\frac{3x+4}{1}, \frac{3y-6}{1} \right)$$

$$\Rightarrow \left. \begin{aligned} x &= 5 \\ y &= -3 \end{aligned} \right\}$$

$$\begin{aligned} \text{(ii)} \quad \int_0^1 \left(\frac{x^2+2}{x^2+1} \right) dx &= \int_0^1 \left(\frac{x^2+1}{x^2+1} + \frac{1}{x^2+1} \right) dx \\ &= \int_0^1 \left(1 + \frac{1}{x^2+1} \right) dx \\ &= \left[x + \tan^{-1} x \right]_0^1 \\ &= (1 + \tan^{-1} 1) - (0 + 0) \\ &= 1 + \frac{\pi}{4} \end{aligned}$$

QUESTION 2: (12 MARKS)

(a) Let roots be α, α, β

$$\text{(i)} \quad 2\alpha + \beta = 0 \quad \text{--- ①}$$

$$\alpha^2 \beta = -2 \quad \text{--- ②}$$

(ii) from ①, $\beta = -2\alpha$ sub in ②

$$\alpha^2 \times (-2\alpha) = -2$$

$$\alpha^3 = 1$$

$$\left. \begin{aligned} \alpha &= 1 \\ \therefore \beta &= -2 \end{aligned} \right\}$$

ie Roots are 1, 1, -2

$$\text{now } \Sigma \alpha \beta \Rightarrow \alpha^2 + 2\alpha\beta = -m$$

$$\text{ie } 1 - 4 = -m$$

$$\therefore m = 3$$

(b) Let roots be $\alpha-d, \alpha, \alpha+d$ --- 3 MARKS

$$\Sigma \alpha \Rightarrow (\alpha-d) + \alpha + (\alpha+d) = \frac{36}{8}$$

$$3\alpha = \frac{36}{8}$$

$$\alpha = \frac{3}{2}$$

$$\Sigma 2B \Rightarrow \frac{3}{2}(\frac{3}{2} - d) + (\frac{3}{2} - d)(\frac{3}{2} + d) = \frac{11}{4}$$

$$\frac{9}{4} - \frac{3d}{2} + \frac{9}{4} - d^2 + \frac{9}{4} + \frac{3}{2}d = \frac{11}{4}$$

$$\text{ie } \frac{27}{4} - d^2 = \frac{11}{4}$$

$$27 - 4d^2 = 11$$

$$4d^2 = 16$$

$$d^2 = 4$$

$$d = 2, -2$$

$$\left. \begin{array}{l} d=2 \Rightarrow \text{roots are } -\frac{1}{2}, \frac{3}{2}, \frac{7}{2} \\ d=-2 \Rightarrow \text{roots are } \frac{7}{2}, \frac{3}{2}, -\frac{1}{2} \end{array} \right\} \text{ie Roots are } -\frac{1}{2}, \frac{3}{2}, \frac{7}{2}$$

(c) (i) $x'' = 4ay$ 3 MARKS.
 $\Rightarrow y = \frac{x''}{4a}$

$$\frac{dy}{dx} = \frac{x}{2a}$$

at $P(2ap, ap^2)$: $\frac{dy}{dx} = \frac{2ap}{2a}$
 $= p$

(ii) PQ through focus $(0, a)$

$$\Rightarrow a = 0 - apq$$

$$\text{ie } pq = -1$$

normal at Q has gradient $-\frac{1}{p}$

tangent at Q has gradient p

$\therefore pq = -1 \Rightarrow$ tangent at P $\perp$ tangent at Q
 ie tangent at P $\parallel$ normal at Q

(d) (i) $f: y = 2^x$
 $f^{-1}: x = 2^y$

$$\therefore \log_2 x = y$$

D: all real x

R: $y > 0$

ie $y = \log_2 x$

(ii) D: $x > 0$

D: $x > 0$

R: all real y

QUESTION 3: (12 MARKS)

(a) $(x - \frac{2}{x^3})^{12}$: $T_{k+1} = {}^{12}C_k x^{12-k} \left(-\frac{2}{x^3}\right)^k$
 $= {}^{12}C_k x^{12-k} (-2)^k x^{-3k}$
 $= {}^{12}C_k (-2)^k x^{12-4k}$

Independent of $x \Rightarrow 12 - 4k = 0$
 $k = 3$

$\therefore$ Term is $T_4 = {}^{12}C_3 (-2)^3$
 $= -1760$ 3 MARKS.

(b) $(2+3x)^{14}$: $T_{k+1} = {}^{14}C_k 2^{14-k} (3x)^k$
 $\therefore T_k = {}^{14}C_{k-1} 2^{14-(k-1)} (3x)^{k-1}$
 $= {}^{14}C_{k-1} 2^{15-k} (3x)^{k-1}$

Then co-efficients P_k, P_{k+1}

$$\begin{aligned} \Rightarrow \frac{P_{k+1}}{P_k} &= \frac{{}^{14}C_k 2^{14-k} 3^k}{{}^{14}C_{k-1} 2^{15-k} 3^{k-1}} \\ &= \frac{14!}{(14-k)! k!} \times \frac{(15-k)! (k-1)!}{14!} \times \frac{3}{2} \\ &= \frac{(15-k) \times 3}{2k} \\ &= \frac{45-3k}{2k} \end{aligned}$$

$$\text{Then } \frac{P_{k+1}}{P_k} > 1 \Rightarrow P_{k+1} > P_k$$

$$\therefore \text{The... } \frac{45-3k}{36} > 1 \Rightarrow 45 > 5k$$

$$k < 9 \Rightarrow P_{k+1} > P_k$$

$$\text{ie } P_9 > P_8 \dots P_3 > P_2 > P_1$$

$\therefore P_9$ is largest co-efficient

$$\text{and } P_9 = {}^{14}C_9 \cdot 2^6 \cdot 3^8$$

$$= 1260971712 \quad \text{--- 4 MARKS.}$$

$$\begin{aligned} \text{(c) (i) } 4 \cos\left(x - \frac{\pi}{3}\right) &= 4\left[\cos x \cos \frac{\pi}{3} + \sin x \sin \frac{\pi}{3}\right] \\ &= 4\left[\cos x \cdot \frac{1}{2} + \sin x \cdot \frac{\sqrt{3}}{2}\right] \\ &= 2 \cos x + \sqrt{12} \sin x \end{aligned}$$

$$\text{(ii) } \sqrt{12} \sin x + 2 \cos x = -2\sqrt{2}$$

$$\Rightarrow 4 \cos\left(x - \frac{\pi}{3}\right) = -2\sqrt{2}$$

$$\cos\left(x - \frac{\pi}{3}\right) = \frac{-\sqrt{2}}{2}$$

$$\begin{aligned} 0 \leq x \leq 2\pi \\ \Rightarrow -\frac{\pi}{3} \leq x - \frac{\pi}{3} \leq \frac{5\pi}{3} \end{aligned}$$

$$\therefore x - \frac{\pi}{3} = \frac{3\pi}{4}, \frac{5\pi}{4}$$

$$\therefore x = \frac{13\pi}{12}, \frac{19\pi}{12}$$

$$= 3.403\dots, 4.974\dots$$

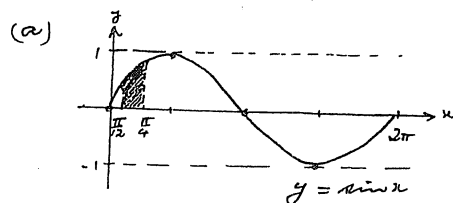
$$= 3.40, 4.97 \text{ (correct to 2 dec places)}$$

$$\left(x - \frac{\pi}{3}\right)_{\text{lower}} = \frac{\pi}{4}$$

$$\begin{array}{c} \checkmark \\ \hline \checkmark \end{array}$$

QUESTION 4: (12 MARKS)

--- 4 MARKS



$$\begin{aligned} V &= \pi \int_{\frac{\pi}{12}}^{\frac{\pi}{4}} y^2 dx \\ &= \pi \int_{\frac{\pi}{12}}^{\frac{\pi}{4}} \sin^2 x dx \\ &= \frac{\pi}{2} \int_{\frac{\pi}{12}}^{\frac{\pi}{4}} (1 - \cos 2x) dx \\ &= \frac{\pi}{2} \left[x - \frac{1}{2} \sin 2x \right]_{\frac{\pi}{12}}^{\frac{\pi}{4}} \\ &= \frac{\pi}{2} \left[\left(\frac{\pi}{4} - \frac{1}{2} \right) - \left(\frac{\pi}{12} - \frac{1}{2} \cdot \frac{1}{2} \right) \right] \\ &= \frac{\pi}{2} \left[\frac{\pi}{6} - \frac{1}{4} \right] \end{aligned}$$

$$\therefore \text{Volume is } \frac{\pi}{2} \left(\frac{\pi}{6} - \frac{1}{4} \right) \text{ units}^3$$

(b) Proposition: $7^n + 5$ is divisible by 6 for all integers $n \geq 1$

$$\text{(i) Test for } n=1: 7^1 + 5 = 12$$

$\therefore$ True for $n=1$

(ii) Assume proposition is true for some integer $n=k$

$$\text{ie } 7^k + 5 = 6M, \text{ } M \text{ integer} \quad \text{--- ①}$$

$$\begin{aligned} \text{Then } 7^{k+1} + 5 &= 7(7^k) + 5 \\ &= 7(6M - 5) + 5 \text{ from ①} \\ &= 42M - 30 \\ &= 6(7M - 5) \\ &= 6N \quad N \text{ integer} \end{aligned}$$

$\therefore$ If proposition is true for $n=k$ then it is also true for $n=k+1$

But true for $n=1 \Rightarrow$ true for 
and hence by the Principle of Mathematical
Induction it is true for all integers $n \geq 1$.

$$(c) \frac{dV}{dt} = 24 \text{ cm}^3/\text{s}$$

$$V = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \pi h^3$$

$$\frac{dV}{dh} = \pi h^2$$

$$\frac{dh}{dt} = \frac{dV}{dV} \cdot \frac{dV}{dt}$$

$$= \frac{1}{\pi h^2} \cdot 24 \text{ cm/s}$$

when $h = 16$

$$\frac{dh}{dt} = \frac{1}{\pi \cdot 16^2} \cdot 24 \text{ cm/s}$$

$$= \frac{3}{32\pi} \text{ cm/s}$$

$$\frac{dS}{dt} = \frac{dS}{dh} \cdot \frac{dh}{dt}$$

$$= 2\pi h \times \frac{3}{32\pi} \text{ cm}^2/\text{s}$$

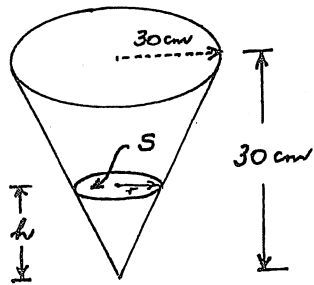
where $S = \pi r^2$
 $= \pi h^2$
 $\frac{dS}{dh} = 2\pi h$

$$h=16 \Rightarrow \frac{dS}{dt} = 3 \text{ cm}^2/\text{s}$$

OR $\frac{dS}{dt} = \frac{dS}{dh} \times \frac{dh}{dV} \times \frac{dV}{dt}$
 $= 2\pi h \times \frac{1}{\pi h^2} \times 24 \text{ cm}^2/\text{s}$

$$h=16 \Rightarrow \frac{dS}{dt} = 32\pi \times \frac{1}{256\pi} \times 24 \text{ cm}^2/\text{s}$$

$$= 3 \text{ cm}^2/\text{s}$$



By similar triangles
 $\frac{r}{h} = \frac{30}{30}$
 $\therefore r = h$

4 MARKS

QUESTION 5 (12 MARKS)

(a)

$$\frac{dT}{dt} = k(T-R) \quad \text{--- 1 MARK}$$

$$(i) T = R + Ae^{kt} \quad \text{--- 2 MARKS}$$

$$\Rightarrow \frac{dT}{dt} = kAe^{kt}$$

$$= k(T-R) \text{ from (2) --- 1 MARK}$$

$$(ii) \frac{dT}{dt} = k(T - -40)$$

$$= k(T + 40)$$

$$\Rightarrow T = -40 + Ae^{kt}$$

$$\left. \begin{matrix} t=0 \\ T=24 \end{matrix} \right\} \Rightarrow 24 = -40 + A$$

$$\therefore A = 64$$

$$\therefore T = -40 + 64e^{kt}$$

$$\left. \begin{matrix} t=5 \\ T=19 \end{matrix} \right\} \Rightarrow 19 = -40 + 64e^{5k}$$

$$59 = 64e^{5k}$$

$$\frac{59}{64} = e^{5k}$$

$$5k = \ln\left(\frac{59}{64}\right)$$

$$\therefore k = \frac{1}{5} \ln\left(\frac{59}{64}\right)$$

$$= -0.016269...$$

$$T=0 \Rightarrow 0 = -40 + 64e^{kt}$$

$$40 = 64e^{kt}$$

$$e^{kt} = \frac{5}{8}$$

$$kt = \ln\left(\frac{5}{8}\right)$$

$$t = \frac{\ln\left(\frac{5}{8}\right)}{k}$$

$$= 28.889...$$

$$= 29 \text{ (to nearest whole)}$$

$\therefore$ It will take 29 s.

(b) $y = \log\left(\frac{2x}{2+x}\right)$

(i) must have $\frac{2x}{2+x} > 0$
 $2x(2+x) > 0$
 $\therefore x < -2$ or $x > 0$ — 2 MARKS
 $\therefore$ Domain is $x < -2$ or $x > 0$

(ii) $y = 0 \Rightarrow \log\left(\frac{2x}{2+x}\right) = 0$
 $\frac{2x}{2+x} = 1$
 $2x = 2+x$ — 1 MARK
 $\therefore x = 2$

(iii) $y = \log\left(\frac{2x}{2+x}\right)$
 $= \log 2x - \log(2+x)$
 $\frac{dy}{dx} = \frac{1}{x} - \frac{1}{2+x}$
 $= \frac{2+x-x}{x(2+x)}$
 $= \frac{2}{x(2+x)}$ — 2 MARKS
 now $2x(2+x) > 0$
 $\therefore x(2+x) > 0$
 $\therefore \frac{dy}{dx} > 0$ for all x in D.
 $\therefore$ fn is increasing

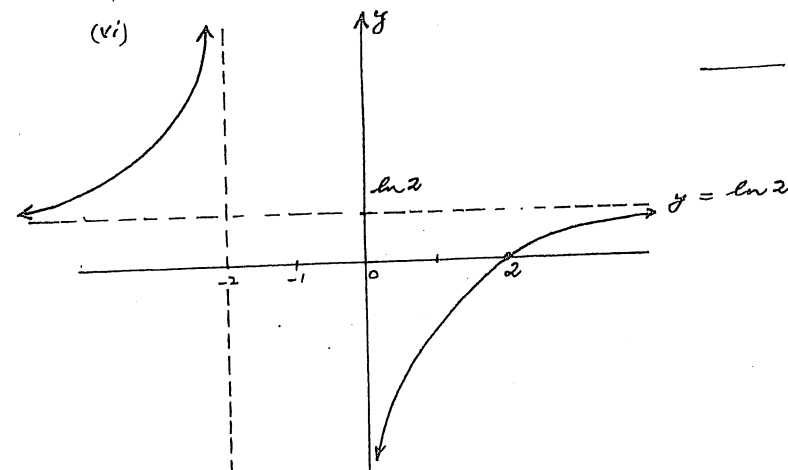
(iv) $\frac{dy}{dx} = \frac{2}{2x+x^2}$
 $\frac{d^2y}{dx^2} = \frac{(2x+x^2) \cdot 0 - 2(2+2x)}{(2x+x^2)^2}$
 $= \frac{-4-4x}{(2x+x^2)^2}$

Possible pt of inflexion when $y'' = 0$
 $-4-4x = 0$
 $x = -1$

But this is outside the domain

$\therefore$ There are no points of inflexion — 1 MARK

(v) $\lim_{x \rightarrow \infty} \left[\log\left(\frac{2x}{2+x}\right) \right]$
 $= \lim_{x \rightarrow \infty} \left[\log\left(\frac{2}{\frac{2}{x}+1}\right) \right]$
 $= \log 2$ — 1 MARK



QUESTION 6: (12 MARKS)

(a) $(1+x)^n = \sum_{r=0}^n {}^nC_r x^r$ — ①

(i) sub $x = 1$ in ①
 $\Rightarrow 2^n = \sum_{r=0}^n {}^nC_r$ — 1 MARK

(ii) now $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_k x^k + \dots + {}^nC_n x^n$
 Differentiating with respect to x :
 $n(1+x)^{n-1} = {}^nC_1 + 2 {}^nC_2 x + \dots + k {}^nC_k x^{k-1} + \dots + n {}^nC_n x^{n-1}$
 $= \sum_{r=1}^n {}^nC_r \cdot r \cdot x^{r-1}$

let $x = 1$
 $\Rightarrow n 2^{n-1} = \sum_{r=1}^n r \cdot {}^nC_r$ — 2 MARKS

$$(b) \int_1^{\sqrt{3}} \frac{dx}{(1+x)\sqrt{x}}$$

$$= \int_1^{\sqrt{3}} \frac{2du}{1+u^2}$$

$$= 2[\tan^{-1}u]_1^{\sqrt{3}}$$

$$= 2[\tan^{-1}\sqrt{3} - \tan^{-1}1]$$

$$= 2\left[\frac{\pi}{3} - \frac{\pi}{4}\right]$$

$$= \frac{\pi}{6}$$

$$u = \sqrt{x}$$

$$= x^{\frac{1}{2}}$$

$$\frac{du}{dx} = \frac{1}{2}x^{-\frac{1}{2}}$$

$$du = \frac{dx}{2\sqrt{x}}$$

$$(ii) \text{ when } x = 6$$

$$= 10 \cos \pi t$$

$$0.6 = \cos \pi t$$

$$(\pi t)_{\text{acute}} = 0.927...$$

$$\text{First time} \Rightarrow \therefore \pi t = 0.92729...$$

$$\therefore t = \frac{0.92729...}{\pi}$$

$$= 0.2951...$$

$$= 0.30 \text{ (correct to 2 dec pl)}$$

$$\therefore \text{Time taken is } 0.30 \text{ s}$$

QUESTION 7:

— 5 MARKS

$$(i) \quad P: \quad \ddot{x}_p = 0 \quad \text{--- (1)}$$

$$\dot{x} = C$$

$$\text{at } t=0, \dot{x}_p = V_p \cos \alpha$$

$$\therefore C = V_p \cos \alpha$$

$$\therefore \dot{x}_p = V_p \cos \alpha \quad \text{--- (2)}$$

$$\Rightarrow x = (V_p \cos \alpha)t + C$$

$$\text{at } t=0, x=0 \therefore C=0$$

$$\therefore x_p = (V_p \cos \alpha)t \quad \text{--- (3)}$$

$$\ddot{y}_p = -g \quad \text{--- (4)}$$

$$\dot{y} = -gt + C$$

$$\text{at } t=0, \dot{y} = V_p \sin \alpha$$

$$\therefore C = V_p \sin \alpha$$

$$\therefore \dot{y}_p = -gt + V_p \sin \alpha \quad \text{--- (5)}$$

$$\Rightarrow y = -\frac{1}{2}gt^2 + (V_p \sin \alpha)t + C$$

$$\text{at } t=0, y=h \therefore C=h$$

$$\therefore y_p = -\frac{1}{2}gt^2 + (V_p \sin \alpha)t + h \quad \text{--- (6)}$$

$$(ii) \quad x_a = (V_a \cos \beta)t$$

$$y_a = -\frac{1}{2}gt^2 + (V_a \sin \beta)t$$

$$(iii) \quad \text{at collision, } t=T, \quad x_p + x_a = 2h \text{ and } y_p = y_a$$

$$\text{now } y_p = y_a$$

$$\Rightarrow -\frac{1}{2}gT^2 + (V_a \sin \beta)T = -\frac{1}{2}gT^2 + (V_p \sin \alpha)T + h$$

$$\therefore T = \frac{h}{V_a \sin \beta - V_p \sin \alpha}$$

$$(c) \quad T = \frac{2\pi}{\omega} = 2$$

$$\therefore \omega = \pi$$

$$\left. \begin{array}{l} t=0 \\ v=0 \\ x=10 \end{array} \right\}$$

(i) The motion can be described by

$$x = 10 \cos \pi t$$

$$\dot{x} = -10\pi \sin \pi t$$

$$(\dot{x})^2 = 100\pi^2 \sin^2 \pi t$$

$$= \pi^2 (100 \sin^2 \pi t)$$

$$\text{ie } v^2 = \pi^2 (100 - x^2)$$

$$\text{when } x=6$$

$$v^2 = \pi^2 \times 64$$

$$v = \pm 8\pi$$

$$\therefore \text{Speed is } 8\pi \text{ m/s}$$

$$v^2 = \omega^2 (a^2 - (x-x_0)^2)$$

$$v^2 = \pi^2 (100 - x^2)$$

Then $x_p + x_a = 2h$

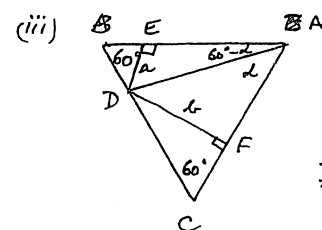
$$\Rightarrow (V_p \cos \alpha) \cdot \frac{h}{V_a \sin \beta - V_p \sin \alpha} + (V_a \cos \beta) \cdot \frac{h}{V_a \sin \beta - V_p \sin \alpha} = 2h$$

$$\therefore V_p \cos \alpha + V_a \cos \beta = 2V_a \sin \beta - 2V_p \sin \alpha$$

$$V_p (\cos \alpha + 2 \sin \alpha) = V_a (2 \sin \beta - \cos \beta)$$

$$\therefore \frac{V_p}{V_a} = \frac{2 \sin \beta - \cos \beta}{2 \sin \alpha + \cos \alpha}$$

— 6 MARKS



$$\text{In } \triangle AED \quad \sin(60^\circ - \alpha) = \frac{a}{AD} = a$$

$$\text{In } \triangle AFD \quad \sin \alpha = \frac{b}{AD} = b$$

From (ii) above,

$$c^2 + cd + d^2 = \frac{3}{4}$$

$$\Rightarrow \frac{a^2}{(AD)^2} + \left(\frac{a}{AD}\right) \cdot \left(\frac{b}{AD}\right) + \frac{b^2}{(AD)^2} = \frac{3}{4}$$

$$a^2 + ab + b^2 = \frac{3}{4} (AD)^2$$

$$\therefore AD = \frac{\sqrt{3}}{3} (a^2 + ab + b^2)$$

(b) (i) $\sin(x-y) = \sin x \cos y - \cos x \sin y$

(ii) $c^2 + cd + d^2$

$$= \sin^2 \alpha + \sin \alpha [\sin(60^\circ - \alpha)] + [\sin(60^\circ - \alpha)]^2$$

$$= \sin^2 \alpha + \sin \alpha \left[\frac{\sqrt{3}}{2} \cos \alpha - \frac{1}{2} \sin \alpha \right] + \left(\frac{\sqrt{3}}{2} \cos \alpha - \frac{1}{2} \sin \alpha \right)^2$$

$$= \sin^2 \alpha + \frac{\sqrt{3}}{2} \sin \alpha \cos \alpha - \frac{1}{2} \sin^2 \alpha + \frac{3}{4} \cos^2 \alpha - \frac{\sqrt{3}}{2} \sin \alpha \cos \alpha + \frac{1}{4} \sin^2 \alpha$$

$$= \frac{3}{4} \sin^2 \alpha + \frac{3}{4} \cos^2 \alpha$$

$$= \frac{3}{4}$$